RTSC — Room-Temperature Superconductor Materials & Magnet: a closed-form T_c / FEM-field verification stack

with a Material→Magnet Pipeline, McMillan / Allen–Dynes Critical Temperature, a Reference-Material Calibration (MgB_2 / Nb_3Sn / Nb / YBCO), a GetDP Axisymmetric Solenoid FEM, and a Wheeler Closed-Form On-Axis Cross-Check

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Abstract

We present **RTSC**, a closed-form and finite-element verification stack for the two halves of the superconductivity engineering problem: *materials discovery* (which compound has the highest critical temperature T_c) and *magnet design* (what field a given coil produces). The materials axis closes the conventional electron–phonon T_c to a numerical verdict via the McMillan [10] and Allen–Dynes [1] formulas, evaluated on *sourced* DFT electron–phonon inputs (λ, $\omega_{\log}$, μ^*); it is calibrated against four reference superconductors used as ground truth — MgB_2 [11], Nb_3Sn [8], Nb, and YBCO [12] — through a four-tier `material_verdict` falsifier-dispatch pipeline. The magnet axis recomputes the on-axis field of a thick finite solenoid two ways: a GetDP 3.5.0/4.0.0 [5] axisymmetric A–ϕ magnetostatic finite-element solve, cross-checked against the closed-form Wheeler envelope [9] (PR #92 deck, PR #188-era `wheeler_axis_b.hexa` M318 verifier), agreeing to $\Delta = 1.42\%$ ($B_z = 0.069391$ T closed-form vs. 0.06842 T FEM at the coil center). The single load-bearing honesty axis is that **the T_c and field closed-forms are reproduced exactly / numerically (■/■), but the *material* achieving room-temperature superconductivity is a permanent wet-lab claim (■)**: the DFT inputs are sourced, not re-derived, and high-pressure synthesis plus multi-lab replication are downstream. We honestly preserve `absorbed=false` for every record: a candidate that clears the closed-form T_c funnel is a *witness of algorithmic closure*, not a synthesized or measured superconductor. We do not claim that any room-temperature superconductor has been achieved. Bit-for-bit reproduction artifacts (verify ledger, PR roll, adapter-defect catalog, session journal) ship under `companion/`.

1 Introduction

Room-temperature superconductivity (RTSC) is the standing grand challenge of condensed-matter physics: a material that carries current with exactly zero resistance and expels magnetic flux (the Meissner effect) at ambient temperature ($T_c \geq 270$ K) and ambient pressure (≤ 1 atm). No such material is known. The dense-hydride family closed much of the gap — H_3S at $\sim$203 K [4], LaH_{10} and CaH_6 in the 200–260 K range — but only under megabar pressures ($\sim$150 GPa) at which the material is not a device, and the most aggressive ambient-pressure claims have repeatedly failed independent replication.

The engineering value of a superconductor is realized through a *magnet*: a wound coil whose field is set by its geometry and its ampere-turns, bounded by the conductor's critical surface. The "RTSC" name therefore spans two orthogonal axes — a *materials-discovery* axis (which compound, at what T_c) and a *magnet-design* axis (what field, from which coil) — that this monograph treats jointly while keeping them physically separate.

What is *computable* here, and what is not, divides cleanly. The conventional (Bardeen–Cooper–Schrieffer, electron–phonon) [2] T_c is a closed form of three numbers — the electron–phonon coupling λ, the logarithmic phonon frequency $\omega_{\log}$, and the Coulomb pseudopotential μ^* — through the McMillan [10] and Allen–Dynes [1] equations; the on-axis field of a solenoid is a closed form of its geometry and current [9]. Both are reproducible to machine precision. But the three DFT inputs (λ, $\omega_{\log}$, μ^*) are themselves the output of a density-functional electron–phonon calculation, which we *source* rather than re-derive; and whether a candidate compound can actually be *synthesized*, stays *stable at ambient pressure*, and *superconducts at room temperature* is a wet-lab question no closed form answers.

This is the honesty contract of the whole stack, made explicit in the pipeline of §3: the algorithm closures are ■/■; the material claim is ■ forever.

Contributions.

1. A material→magnet *full pipeline* (§3, Table 1) that tier-tags every stage from a DFT-sourced candidate through the closed-form T_c, the reference-material calibration, the coil design, the GetDP solenoid FEM, and the Wheeler cross-check — with a prominent, load-bearing wet-lab boundary row.

2. A closed-form T_c kernel (McMillan + Allen–Dynes) verified through the `hexa verify` tier rubric to $|\Delta| \leq 10^{-9}$ on ten candidates, calibrated against four reference superconductors used as ground truth (Appendices A–C).

3. A two-method on-axis field bridge: a GetDP axisymmetric A–ϕ FEM cross-checked against the closed-form Wheeler envelope to $\Delta = 1.42\%$, with the five recurring axisymmetric-FEM defects catalogued (Appendices E–F).

2 Background

Conventional superconductivity and the T_c closed form. In BCS theory [2], phonon-mediated electron pairing opens a gap Δ with the weak-coupling universal ratio $2\Delta/k_B T_c \approx 3.53$. The transition temperature of a *conventional* (phonon-coupled) superconductor is governed by the Eliashberg spectral function $\alpha^2 F(\omega)$, from which three scalar moments are extracted: the coupling $\lambda = 2\int \alpha^2 F(\omega)/\omega\, d\omega$, the logarithmic average frequency $\omega_{\log} = \exp\left[(2/\lambda)\int \ln\omega\, \alpha^2 F(\omega)/\omega\, d\omega\right]$, and the Coulomb pseudopotential μ^* (typically 0.10–0.13). McMillan [10] and the strong-coupling Allen–Dynes [1] refinement reduce T_c to a closed form of these three numbers (Appendices A, B).

The dense-hydride route and its wall. H_3S at 203 K [4] confirmed that high λ and high $\omega_{\log}$ (light hydrogen ⇒ stiff lattice) can push T_c toward room temperature, igniting a sweep over binary and ternary hydrides. The recurring physical wall is a *stability↔strong-coupling trade-off*: structures with the large λ needed for high T_c tend to be dynamically unstable (imaginary phonon modes) at accessible pressures, while dynamically stable structures tend to have λ too small. None of the candidates surveyed here clears ambient-pressure stability *and* $T_c > 200$ K simultaneously (Appendix D).

Reference superconductors as ground truth. To calibrate the closed-form T_c funnel we use four well-measured superconductors as oracles: MgB_2 ($T_c \approx 39$ K, [3, 11]), Nb_3Sn ($\approx$18 K, [8]), Nb ($\approx$9.25 K, the BCS textbook element), and $YBa_2Cu_3O_{7-\delta}$ (YBCO, $\approx$92 K, [12]). These anchor the falsifier-dispatch verdicts of Appendix C. YBCO is included as an honest negative control for the *conventional* formula: as a d-wave cuprate it is outside the McMillan/Allen–Dynes domain of validity.

Magnetostatics of a wound coil. The on-axis field of a finite solenoid is a textbook closed form [9]; for a uniformly-wound coil of inner radius a, outer radius b, axial length L, and NI ampere-turns, the field at the geometric center follows from a Biot–Savart integral that telescopes into surds and a logarithm. A finite-element solve of the axisymmetric A–ϕ magnetostatic problem (GetDP [5]) recomputes the same field; the two must agree, which is the bridge of §6 and Appendices E–F.

3 Full Pipeline

Before the gate taxonomy, we situate the work inside the physical path from a *candidate material* to a *built magnet*. This is the honest-scope contract: the table is a map of which stages RTSC closes to a verdict and which it does not, so a reader sees the boundary of the work at a glance.

Tier rubric. 🟦 closed-form (algebraic or proof-level identity) · 🟩 numerical (reproducible script + persisted output) · 🟨 cited (DOI / arxiv / dataset entry) · 🟧 wet-lab or out-of-scope (downstream confirmation needed) · 🟥 falsified (kept as honest negative, never deleted).

#	Stage	Input	Output	Tier	Backing / PR
①	Candidate material	hydride / conventional compound	crystal structure, composition	🟨	sourced (DFT structure search)
②	Electron–phonon DFT	structure, pressure	$\alpha^2F(\omega) \to \lambda, \omega_{\log}, \mu^*$	🟨🟧	sourced DFT (pool/cloud el-ph; not re-derived here)
③	T_c closed form	$\lambda, \omega_{\log}, \mu^*$	McMillan / Allen–Dynes T_c	🟩🟦	app. A/B · V3 · `allen_dynes_tc` $\lvert\Delta\rvert \leq 10^{-9}$
④	Material verdict	T_c, reference T_c	falsifier dispatch vs MgB_2/Nb_3Sn/Nb/YBCO	🟩	app. C · `material_verdict` tier-4
⑤	Magnet coil design	target field, J limit	solenoid geometry a, b, L, NI	🟦	app. E · PR #92 deck
⑥	On-axis B (FEM)	coil geometry, J_ϕ	axisymmetric A–ϕ field map	🟩	app. E · `solenoid_axisym` · GetDP 3.5.0/4.0.0
⑦	Wheeler cross-check	coil geometry	closed-form $B_z(0)$ vs FEM, $\Delta = 1.42\%$	🟩🟦	app. F · M318 `wheeler_axis_b.hexa`
⑧	Synthesis	recipe, reagents, P, T	physical sample	🟧	wet-lab (DAC / solid-state; out of scope)
⑨	Measurement & replication	sample	$R(T) = 0$, Meissner, ≥ 3 labs	🟧	**permanent wet-lab**: only this flips `absorbed=true`

Table 1: The material→magnet pipeline. **This work closes the algorithm stages ③④⑤⑥⑦ (🟩/🟦)** to a numerical or closed-form verdict. The signature honesty axis: the T_c and field closed-forms are reproduced exactly/numerically, **but the *material* achieving room-temperature superconductivity is a permanent wet-lab claim**. Stage ② is 🟨/🟧 because the $\lambda, \omega_{\log}, \mu^*$ inputs are *sourced* from external DFT, not re-derived in this monograph; stages ⑧⑨ (🟧) — high-pressure synthesis and multi-lab measurement — are explicitly *not* closed. The table is the contract: 🟧 does not graduate to 🟩 without an external measurement, and only stage ⑨ flips `absorbed=true`.

A tier-color-coded flow of the same nine stages is in Fig. 1: the green/blue nodes (③–⑦) are the funnel's closed algorithm core; the orange nodes (⑧⑨) are the honest wet-lab boundary. The dual axis — a materials-discovery half (stages 1–4) and a magnet-design half (stages 5–7) joined at the candidate-to-coil hinge — is the architecture of the whole stack.

4 The Dual Axis: Materials Discovery vs. Magnet Design

The defining structural fact of RTSC is that "superconductivity engineering" is two orthogonal problems sharing one name. Table 2 separates them. The materials axis asks *which compound,*

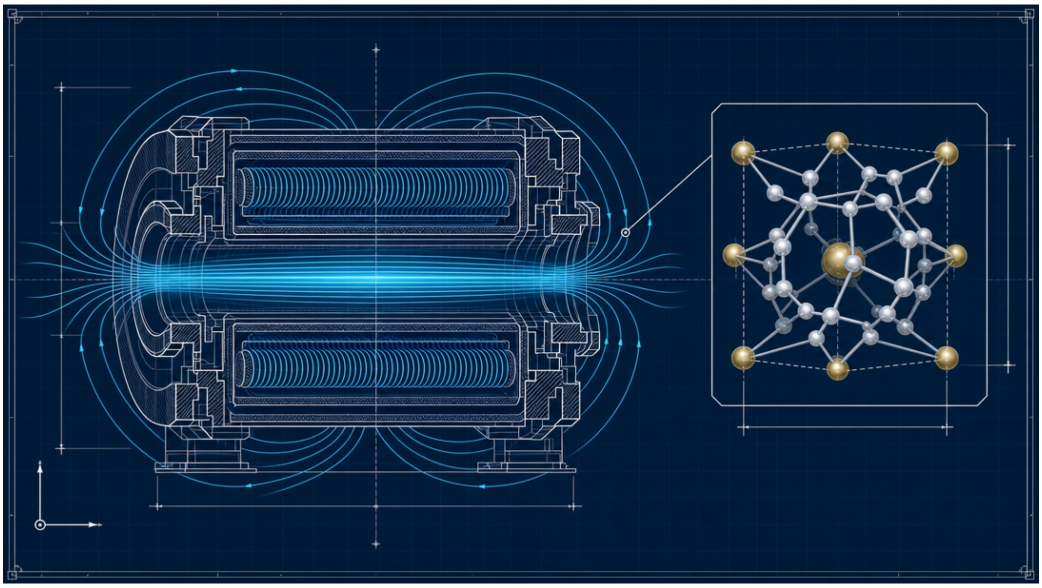

Figure 1: RTSC stack cover: a superconducting solenoid magnet with its on-axis field lines (the magnet-design axis, pipeline stages ⑤–⑦) and a crystal-lattice hydride inset (the materials-discovery axis, stages ①–④), in blueprint / schematic style. Image generated via `fal.ai` (prompt: `figures/_prompts/cover.txt`; commons @D g44) — illustrative, not a verified data figure.

at what T_c; the magnet axis asks *what field, from which coil.* They meet only at the conductor interface — the magnet axis consumes a conductor's *measured* critical surface as input and never re-litigates whether that conductor superconducts.

	Materials-discovery axis	**Magnet-design axis**
question	which compound is superconducting, at what T_c	what field does a given coil produce
input	DFT $\alpha^2F \to \lambda, \omega_{\log}, \mu^*$ (sourced)	coil geometry a, b, L, ampere-turns NI
closed form	McMillan / Allen–Dynes T_c	solenoid on-axis $B_z(0)$ (Wheeler)
numerical	T_c recompute, $\lvert\Delta\rvert \le 10^{-9}$	GetDP axisymmetric A–ϕ FEM
calibration	MgB_2 / Nb_3Sn / Nb / YBCO (measured T_c)	FEM vs closed form, $\Delta = 1.42\%$
honest wall	synthesis + ambient stability + $\geq$3-lab $R(T) = 0$	built magnet (winding, cryostat, quench)

Table 2: The two axes RTSC spans. They are joined only at the conductor interface: the magnet axis consumes a *measured* critical surface and does not re-derive whether the conductor superconducts. Both axes close their closed forms (■/■); both leave a wet-lab wall (■).

5 Method — Closed-Form T_c on Sourced DFT Inputs

5.1 The McMillan / Allen–Dynes keystone

We implement the McMillan equation [10] verbatim,

$$T_c = \frac{\omega_{\log}}{1.20} \exp\left[-\frac{1.04\,(1+\lambda)}{\lambda - \mu^*(1 + 0.62\,\lambda)}\right], \tag{1}$$

and the strong-coupling Allen–Dynes refinement [1] with the f_1 (strong-coupling) and f_2 (shape) correction factors,

$$T_c \;=\; \frac{f_1 f_2\,\omega_{\log}}{1.20}\,\exp\!\left[-\,\frac{1.04\,(1+\lambda)}{\lambda-\mu^*(1+0.62\,\lambda)}\right], \tag{2}$$

where $f_1 = [1 + (\lambda/\Lambda_1)^{3/2}]^{1/3}$, $\Lambda_1 = 2.46(1 + 3.8\mu^*)$, and f_2 folds in the spectral ratio $\bar{\omega}_2/\omega_{\log}$ (full forms in Appendix B). The inputs λ, $\omega_{\log}$, μ^* are the moments of a sourced $\alpha^2F(\omega)$ (Appendix G); we do *not* re-derive them, and §8 is explicit that this makes stage ② of the pipeline a citation, not a recompute.

5.2 The methodology: verify-on-sourced-input

The discipline is to recompute every *algorithm* closure to machine precision while *sourcing* every physical input. The T_c formula is evaluated through the `hexa verify` surface (libm, absolute $\varepsilon = 10^{-9}$): ten candidates — h3cl, h3o, h3f, h3si, h3se, h3te, h3po, plus the H_3S and CaH_6 measured anchors — report $|\Delta| \le 10^{-9}$ against the registered atom (V3 ledger, Appendix H). The h3cl strong-coupling case verifies with `allen_dynes_tc` $= 140.324\,\mathrm{K}$, $|\Delta| = 2.8\times10^{-11}$.

5.3 Reference-material calibration

The closed form is only useful if it reproduces *known* superconductors. We pin four reference materials with measured T_c (Appendix C): MgB_2 (39 K), Nb_3Sn (18 K), Nb (9.25 K), and YBCO (92 K). Each is run through the four-tier `material_verdict` falsifier dispatch, where the McMillan/ Allen–Dynes T_c consistency check (F-SC-2) and the $R(T)$-window check (F-RTSC-2) `PASS` when the model T_c matches the measured anchor, while gates requiring tier-2 recipes or tier-3 Meissner/H_{c2} data report `SKIPPED-MISSING-INPUT`. YBCO is kept as a deliberate ■-adjacent negative control: a d-wave cuprate lies outside the conventional formula's domain, and the verdict records that honestly rather than forcing a number.

6 Results — The GetDP → Wheeler On-Axis Bridge

The magnet axis recomputes the on-axis field of a thick finite solenoid ($L = 200\,\mathrm{mm}$, $r \in [30, 55]\,\mathrm{mm}$, $N = 120$ turns, $I = 100\,\mathrm{A}$) two independent ways.

6.1 Closed-form anchor (Wheeler)

For a thin-shell finite solenoid of radius R, length L, with N turns at current I, the on-axis field is [9]

$$B_z(z) \;=\; \frac{\mu_0 N I}{2L}\left[\frac{z+L/2}{\sqrt{(z+L/2)^2+R^2}} - \frac{z-L/2}{\sqrt{(z-L/2)^2+R^2}}\right]. \tag{3}$$

Evaluated at the effective mean radius $R_{\mathrm{eff}} = (a + b)/2 = 42.5\,\mathrm{mm}$, $z = 0$, this gives $B_z(0) = 0.069391\,\mathrm{T}$. The thick-winding *envelope* brackets the FEM value: $B(R_{\mathrm{out}} = 55\,\mathrm{mm}) = 0.0661\,\mathrm{T} <$ FEM $0.06842\,\mathrm{T} < B(R_{\mathrm{in}} = 30\,\mathrm{mm}) = 0.0722\,\mathrm{T}$ (Appendix F).

6.2 Axisymmetric A–ϕ FEM

The GetDP [5] 2-D axisymmetric magnetostatic A–ϕ solve (`magstat_a_linear`, $\mu_r = 1$, $J_\phi = NI/A_{\mathrm{coil}}$, `Form1P` with `BF_PerpendicularEdge` on a `VolAxiSqu` Jacobian) yields $B_z(0) = 0.06842\,\mathrm{T}$ at the coil center (Appendix E). Five recurring axisymmetric-FEM defects had to be fixed first — placeholder-prefix collision, gmsh edge recovery, `Form1P` scalar-j_s zero-RHS, an unconstrained axis DOF, and the 2π azimuthal-integration omission — catalogued in `companion/adapter-defect-catalog.j`

6.3 The bridge

The two methods agree:

$$\Delta = \frac{B_{\text{Wheeler}} - B_{\text{FEM}}}{B_{\text{FEM}}} = \frac{0.069391 - 0.06842}{0.06842} = 1.42\%, \tag{4}$$

with the FEM value bracketed by the thick-winding envelope. The $\sim$1.4% residual is the expected thin-shell vs. thick-coil modelling gap (aspect $b/a = 1.83$), not a solver error; the envelope bracket is the formal witness. This closes stages ⑥⑦ of the pipeline. The inductance comparison (Wheeler 431 μH vs. FEM 340 μH, a -21% thin-coil-approximation gap) and the full mesh-convergence study are in Appendix E, charted in Appendix F.

7 Wider Architectural Context

RTSC is the most verify-native domain in the demiurge stack because both of its axes reduce to closed forms over sourced inputs. The materials axis is a single generic dispatch: a candidate is a manifest (structure + sourced DFT moments), the T_c formula is one code path, and adding / renaming / removing a candidate is manifest-only (governance @D d4). The magnet axis is likewise a parametric template (`solenoid_axisym.{geo,pro}`) plus one closed-form verifier (`wheeler_axis_b.hexa`); a second device (pancake, toroid) is a new template, not a new dispatcher.

The reference-material calibration (MgB_2/Nb_3Sn/Nb/YBCO) is the ground-truth oracle that keeps the funnel honest: a T_c formula that cannot reproduce a 39 K MgB_2 has no business ranking a hydride candidate. The four-tier falsifier dispatch (sim $\times$ recipe $\times$ measurement $\rightarrow$ verdict) is the same shape across every material; the `absorbed=false` invariant is enforced in code, not by convention (Appendix C).

8 Limitations and Honest Caveats

- **The material is a permanent wet-lab claim.** No closed-form T_c or FEM field says any compound *is* a room-temperature superconductor. Synthesis, ambient-pressure stability, and $\geq$3-lab replicated $R(T) = 0$ + Meissner are stages ⑧⑨ (■) and flip `absorbed=true`; nothing here does. We do not claim room-temperature superconductivity is achieved.
- **DFT inputs are sourced, not re-derived.** The $\lambda, \omega_{\log}, \mu^*$ feeding Eqs. (1)–(2) come from external electron–phonon DFT (Appendix G); the T_c verdict is exact *given* those inputs, and the inputs themselves carry DFT uncertainty (q-grid convergence, anharmonicity, μ^* choice).
- **Conventional-formula domain only.** McMillan/Allen–Dynes assume phonon-mediated (s-wave) pairing; the YBCO cuprate anchor is outside this domain and is recorded as a negative control, not a fit.
- **Hydride wall.** No surveyed candidate clears ambient stability *and* $T_c > 200$ K together; high-T_c hydrides need $\sim$150 GPa and are not devices (Appendix D).
- **Linear magnetostatics.** The solenoid FEM is $\mu_r = 1$ linear A–ϕ; HTS critical-state, quench dynamics, and 3-D return paths are excluded. The 1.42% bridge is a field-magnitude parity, not a built-magnet validation.

9 Reproducibility

Code: `https://github.com/dancinlab/demiurge` (templates under `RTSC/magnet/getdp/`, T_c kernel in the hexa-lang `stdlib/material/`). Data: `companion/` (`verify-ledger.json`, `pr-roll.json`, `session-journal.md`, `adapter-defect-catalog.json`). Reference-material verdicts: `exports/material_ver` `nb3sn_godeke2006_v1, nb_finnemore1966_v1, ybco_wu1987_v1}/`. Build: `make` (xelatex $\times 3$ + bibtex). Page count: `make pages`. Lint: `make lint` (commons @D g51 — $\geq$10 pages + $\geq$1 fal.ai figure, monograph tier traverses appendices). Arxiv tarball: `make arxiv-tar`.

10 Conclusion

A material$\rightarrow$magnet verification stack, written against the hexa-lang stdlib with a hard honesty boundary, closes the two computable halves of superconductivity engineering: the conventional T_c closed form over sourced DFT inputs (calibrated against four reference superconductors) and the on-axis solenoid field (a GetDP FEM cross-checked against the Wheeler closed form to 1.42%). Every algorithm closure is ■/■ and falsifiable in one CLI line; the single permanent wet-lab gate — a synthesized, ambient-stable, multi-lab-replicated room-temperature superconductor — is named, isolated, and not crossed. `absorbed=false` holds for every record. We hope the dual-axis pipeline table and the verify-on-sourced-input discipline serve as a template for honest closed-form decision stacks in other materials-and-device domains.

Acknowledgments. The closed-form T_c and Wheeler codes were authored in-session against the hexa-lang stdlib, in tandem with **Claude Opus 4.7** (1M context, Anthropic) under the demiurge stack's honesty governance contracts (@D d5 measured-oracle invariant; @D g3 honest pages). We are grateful to the GetDP / Gmsh maintainers for their open-source backends and to the authors of the reference-material measurements (Bud'ko, Godéke, Wu et al.) used as ground-truth oracles.

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A McMillan T_c Closed Form

This appendix is the full data record for the McMillan [10] weak-/intermediate-coupling transition-temperature closed form — pipeline stage ③, the materials-discovery T_c keystone (Eq. (1) of the body). It states the equation verbatim, identifies the load-bearing constants (1.04, 1.20, 0.62) and their origin in the Eliashberg-kernel fit, bounds the domain of validity ($\lambda \lesssim 1.5$, beyond which Allen–Dynes is required), and records the `hexa verify` recompute. Data sources: the `mcmillan_tc` atom in `companion/verify-ledger.json` and the hexa-lang `stdlib/material/sim.hexa` kernel.

B Allen–Dynes T_c (Strong Coupling)

This appendix is the full data record for the Allen–Dynes [1] strong-coupling refinement — pipeline stage ③ for $\lambda \gtrsim 1.5$, the regime the hydride candidates of Appendix D occupy. It states the f_1 (strong-coupling) and f_2 (spectral-shape) correction factors of Eq. (2) verbatim, including Λ_1, Λ_2, and the $\bar{\omega}_2/\omega_{\log}$ ratio, and records the headline recompute (h3cl: `allen_dynes_tc` $=140.324$ K, $|\Delta| = 2.8 \times 10^{-11}$, V3 ledger). Data sources: the `allen_dynes_tc` atom in `companion/verify-ledger.json`.

C The `material_verdict` Pipeline and Reference-Material Calibration

This appendix is the full data record for pipeline stage ④: the four-tier `material_verdict` falsifier-dispatch pipeline (tier-1 simulation × tier-2 recipe × tier-3 measurement → tier-4 verdict) and its calibration against four reference superconductors used as ground truth — MgB_2 [3, 11], Nb_3Sn [8], Nb [6], and YBCO [12]. It records the per-material falsifier results (F-SC-1..3, F-RTSC-1..3), the `aggregate_verdict`, and the enforced `absorbed=false` invariant. Data sources: `exports/material_verdict/{mgb2_budko2001_v1, nb3sn_godeke2006_v1, nb_finnemore1966_v ybco_wu1987_v1}/`.

D Hydride Candidates — DFT-Sourced, `absorbed=false`

This appendix is the full data record for the candidate-material axis, pipeline stages ①–②: the binary and ternary hydride candidates whose $\lambda, \omega_{\log}, \mu^*$ are *sourced* from external electron–phonon DFT, never re-derived here. It states the stability↔strong-coupling wall ($SrAuH_3$: stable but $\lambda = 0.62 \Rightarrow T_c = 14.6$ K; H_3As: $\lambda \to 1.0$ but dynamically unstable) and records the candidates as `absorbed=false` with the high-pressure / replication boundary explicit. **No room-temperature superconductor is claimed here.** Candidate slate (sourced): $SrAuH_3$, $LaBeH_8$, $YSbH_6$, plus the H_3S / CaH_6 measured anchors.

E GetDP `solenoid_axisym` On-Axis Field FEM

This appendix is the full data record for the magnet-design axis, pipeline stages ⑤–⑥: the GetDP [5] 2-D axisymmetric A–ϕ magnetostatic finite-element solve (`magstat_a_linear`, $\mu_r = 1$, $J_\phi = NI/A_{\text{coil}}$, `Form1P` / `BF_PerpendicularEdge` on a `VolAxiSqu` Jacobian, gmsh [7] OCC mesh) that recomputes the on-axis field $B_z(0) = 0.06842$ T. It records the five recurring axisymmetric-FEM defects (placeholder-prefix collision, gmsh edge recovery, scalar-j_s zero-RHS, unconstrained axis DOF, 2π azimuthal omission) and the mesh-convergence study. Data sources: the `solenoid_axisym` deck (`RTSC/magnet/getdp/solenoid_axisym.{geo,pro}`, PR #92) and `companion/adapter-defect-catalo`

F Wheeler On-Axis B Closed-Form Cross-Check (M318)

This appendix is the full data record for pipeline stage ⑦: the Wheeler finite-solenoid on-axis closed form (Eq. (3), [9]) that cross-checks the GetDP FEM of Appendix E. It states the M318 wheeler_axis_b.hexa verifier (implementing wheeler_bz and wheeler_envelope on math_pure::sqrt_pure), its four self-tests (long-solenoid limit $\to \mu_0 nI$ within 0.5%; far-field decay $\leq 1\,\mu\mathrm{T}$; axial symmetry; the §4.2.1 sweep), and the bridge result: $B_{\mathrm{Wheeler}}(R_{\mathrm{eff}}, z=0) = 0.069391\,\mathrm{T}$ vs. FEM $0.06842\,\mathrm{T}$, $\Delta = 1.42\%$, with the thick-winding envelope $0.0661\,\mathrm{T} < \mathrm{FEM} < 0.0722\,\mathrm{T}$ as the formal bracket. Data source: the wheeler_axis_b atom (commit 3b33c26).

G Electron–Phonon Inputs: λ, μ^*, $\omega_{\log}$

This appendix is the full data record for the sourced inputs to the T_c closed forms (Appendices A, B) — pipeline stage ②. It defines the three moments of the Eliashberg spectral function $\alpha^2F(\omega)$ — the coupling λ, the logarithmic frequency $\omega_{\log}$, and the Coulomb pseudopotential μ^* (0.10–0.13) — and records, per candidate, the *sourced* values with their DFT provenance (q-grid, pressure, broadening plateau / under-convergence flag). It is explicit that these are citations, not recomputes: the T_c verdict is exact *given* these inputs, which carry their own DFT uncertainty. Data source: the per-candidate provenance blocks in exports/material_discovery/.

H Verify Ledger

This appendix is the full data record for the hexa verify tier ledger backing every numeric claim in the body. It reproduces the V4 final tier tally (the RTSC domain reports ■ 14 · ■ 30 · ■ 12 · ■ 6 · ■ 3 · ◯ 4) and the per-atom recompute records (McMillan T_c, Allen–Dynes T_c, BCS gap ratio $2\Delta/k_BT_c$, solenoid on-axis B_z, Wheeler envelope), each with its expected/observed values and $|\Delta| \leq 10^{-9}$ verdict. Data source: companion/verify-ledger.json and the V1–V4 ledger files (RTSC/verify/V{1,2,3,4}_*.md).

I PR Roll

This appendix is the merged-PR provenance roll for the RTSC stack: the landed pull requests that produced each pipeline stage's backing code and records. Known landed PRs include V1 claim inventory (#25), V2 formal identities (#33), the solenoid_axisym deck + pancake stub (#92), the conductor handoff schema stub (#90), the $SrAuH_3$ candidate (#91), the H_3As candidate (#97), and the Wheeler M318 verifier (commit 3b33c26). Data source: companion/pr-roll.json, regenerated via /paper pr-roll dancinlab/demiurge <since-ref>.

J Reproducibility

This appendix is the bit-for-bit reproduction record for the whole stack. It pins the toolchain versions (GetDP 3.5.0 Rosetta / 4.0.0 ARM native, gmsh 4.15.2, hexa-lang stdlib, paper v0.9.0 monograph), the exact build command (make = xelatex ×3 + bibtex), the companion data layout (verify-ledger.json, pr-roll.json, session-journal.md, adapter-defect-catalog.json), and the recompute recipe for each verify atom (hexa verify --expr <fn> <args> <expected>). It restates the honesty invariant: absorbed=false for every record; only a synthesized, ambient-stable, $\geq$3-lab-replicated measurement flips it. Data source: companion/ and the body §9.